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REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTERE DES ENSEIGNEMENTS SECONDAIRES

CELLULE D'APPUI A L'ACTION PEDAGOGIQUE
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MINISTRY OF SECONDARY EDUCATION

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GCE AL

MARCH 2025

The Teachers' Resource Unit and the Regional Inspectorate of Pedagogy in collaboration with NWCTA	SUBJECT CODE NUMBER 0715	PAPER NUMBER 2
GENERAL CERTIFICATE OF EDUCATION REGIONAL MOCK EXAMINATION	SUBJECT TITLE CHEMISTRY	
ADVANCED LEVEL		

Time Allowed: THREE HOURS

INSTRUCTIONS TO CANDIDATES

Mobile phones are NOT ALLOWED in the examination room.

Answer all SIX questions in the booklet

The mark allocation is indicated for each question. Each question carries 20 marks

Verify that this booklet contains six questions, questions are repeated, and there are no blank pages.

Blank spaces in this booklet may be used for rough work.

In calculations, you are advised to show all your working, giving answers at each stage.

Calculators may be used.

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SECTION A: PHYSICAL AND GENERAL CHEMISTRY

1. (a) Define the following terms as applied to the atom

(i) A molecule

.....

.....

Mass number

.....

.....

(2 marks)

(b) How many protons and neutrons are contained in $^{59}_{27}Co$.

.....

(1 mark)

(c) Considering an atom, when is an electron said to be

(i) In the ground state

.....

(ii) In the excited state

..... (2 marks)

(d) Complete the table below by giving the shapes and bond angles of the species given

Species	Shape	Bond angle
SO_4^{2-}		
NH_2^-		

(2 marks)

(e) (i) Explain the terms acid and conjugate base as used in the Bronsted-Lowry theory of acids and base

Acids:

.....

Conjugate base:

..... (2 marks)

(ii) A 0.010M solution of NH_3 has a pH of 10.6 at 25°C. Calculate the concentration of OH^- ions in solution.

.....

.....

..... (3 marks)

(f) Define the terms below and illustrate using suitable equations.

(i) Standard enthalpy of atomization:

.....

.....

.....

(ii) Standard lattice energy:

.....

..... (4 marks)

(g) (i) Define the term 'average rate' of a reaction.

.....

.....

(ii) During the hydrolysis of an ester, the concentration of the ester decreases from 1.00 mol/dm^3 to 0.030 mol/dm^3 in 1 hour. What is the average rate of the reaction over this time interval?

.....

.....

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(iii) Sketch a graph depicting how the concentration of a reactant changes with time as the reaction progresses.

.....

.....

..... (4 marks)

(Total = 20 marks)

2. (a)(i) What is a standard solution?

.....(1 mark)

(ii) Determine the concentration of a potassium permanganate solution used to titrate 25 cm^3 of 2.388M acidified sodium oxalate ($Na_2C_2O_4$) solution, using 18 cm^3 of the $KMnO_4$ solution. The equation for the reaction is shown below.

$$2MnO^{4-}_{(aq)} + 5C_2O^{2-}_{4\,(aq)} + 16H^{+}_{(aq)} \rightarrow 2Mn^{2+}_{(aq)} + 10CO_{2\,(g)} + 8H_2O_{(l)}$$

.....

..... (2 marks)

(b) X-rays with a wavelength of 1.54Å are incident on a crystal with an inter planar spacing of 3.35Å. Calculate the angle of incidence for the first-order (n=1) using the Bragg equation.

.....

..... (2 marks)

(c) A trichloromethane ($CHCl_3$) and acetone (CH_3COCH_3) mixture forms a non-ideal solution and deviates negatively from Raoul's law. Draw well-labeled vapour pressure-composition and boiling point-composition curves for the mixture.

.....

.....

..... (4 marks)

(d) 0.10 mol of N_2O_4 is placed into a 0.10 dm^3 flask and allowed to reach equilibrium with NO_2 at 398K, according to the equation.

$$N_2O_{4\,(g)} \rightleftharpoons 2NO_{2\,(g)}$$

At equilibrium, there is 0.071 mol of N_2O_4.

(i) Using this system as an example, explain what is meant by "dynamic equilibrium".

..... (1 mark)

((ii)what property of the system must be measured to enable the calculation of the equilibrium constant?

..... (1 mark)

(ii) Write the equilibrium constant expression (Kc) for the system

..... (1 mark)

(iii) Calculate the equilibrium constant, stating units.

.....

.....(3 marks)

e) Consider the galvanic cell consisting of the following half-reactions.

A) $MnO^{-}_{4\,(aq)} + 8H^{+}_{(aq)} + 5e^{-} \rightleftharpoons Mn^{2+}_{(aq)} + 4H_2O_{(l)}$ $\quad E^{\ominus} = +1.52\ V$

B) $Sn^{4+}_{(aq)} + 2e^{-} \rightleftharpoons Sn^{2+}_{(aq)}$ $\quad E^{\ominus} = +0.15\ V$

(i) Write an equation to show the overall reaction in the cell.

..... (1 mark)

(ii) Write the cell diagram and state which will be the positive electrode of the redox system.

..... (2 marks)

(iii) Define the term "standard cell emf.

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.....

.....(1 mark)

(iv) Calculate the emf (E^{θ}_{cell}) of a cell constructed from the above redox systems.

.....(1 mark)

(Total = 20 marks)

SECTION B: INORGANIC CHEMISTRY

3. This question is on periodicity, the halogens, and nitrogen.

(a) The elements in period 2 of the periodic table show periodic trends in their properties.

What is meant by the term "periodic trend"?

.....(1 mark)

(b) (i) What are isoelectronic ions?

.....(1 mark)

(ii) List the isoelectronic anions of period 2 elements.

.....(1 mark)

(iii) State and explain how the ionic radii of the isoelectronic anions vary across the period 2 elements.

Trend:

.....

Explanation:

.....

.....(2 marks)

(c) Which element in period 2 has the highest electronegativity?

.....(1 mark)

(d) Halogens are elements of Group VII of the periodic table and exist as diatomic molecules.

(i) What is a diatomic molecule?

.....

(ii) State the general outer electronic configuration exhibited by halogens.

.....

(iii) Write a balanced equation for the reaction of bromine with cold dilute NaOH.

.....

(iv) State and define the type of reaction taking place in c(iii) above.

Type of reaction:

.....

Definition:

.....

.....(5 marks)

(v) Arrange the oxoacids of chlorine ($HClO$, $HClO_2$, $HClO_3$, $HClO_4$) in terms of decreasing thermal stability.

.....

Explain your reasoning

.....

.....(3 marks)

(e) (i) Draw a dot and cross diagram to illustrate the bonding in the formation of a nitrogen molecule from two nitrogen atoms.

.....

.....(2 marks)

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(ii) Using chemical equations only, outline the different steps involved in the industrial preparation of nitric acid.

.....

..... (3 marks)

(iii) State one large-scale use of nitric acid

.....(1 mark)

(Total = 20 marks)

---

4. (a) Li, Na, K, Rb, Be, Mg, Ca, Ba, and Sr are S block elements.

(i) Write the electronic configuration of potassium and calcium

K: .....

Ca^{2+}: .....

(2 marks)

(ii) In the same period, the boiling points of Group II metals are generally higher than those of Group I metals. Explain.....

.....**(1 mark)**

(iii) State and explain the trend in the solubility of Group I and Group II metal sulphates down each group.

Trend.....

Explanation:

.....

(3 marks)

(iv) Why are the Group II metal hydroxides less soluble in water than the Group I metal hydroxides?

.....

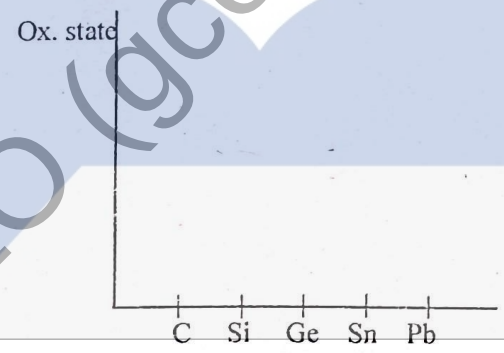
.....**(2 marks)**

(b) This part of the question concerns the Group 4 elements (carbon, Silicon, germanium, tin, and lead).

(i) Explain what the term "inert pair effect" means.

.....

(ii) In the axis below, sketch the trend in the stability of the +2 oxidation state down the Group IV elements.



(iii) Explain the trend indicated above.

.....

(iv) Write equations for the hydrolysis of CCl_4 and $SiCl_4$.

.....

.....**(4 marks)**

(c) Consider the first row of d-block elements (Scandium to zinc)

Which of these elements is/are not transition metals? Explain.

.....

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(d) Transition metals usually form complexes. The cobalt ion Co^{3+} form a complex ion having the formula $[Co(NH_3)_6]^+$ in which ammonia is acting as a monodentate ligand.

(i) What is meant by monodentate ligand? ..... (1 mark)

.....

(ii) Give the systematic name of this complex ion. ..... (1 mark)

.....

(iii) Draw the structure of the complex ion ..... (1 mark)

.....

(e) Two of the ammonia ligands in the above complex are replaced by chloride ligands, an ion of formula $[Co(NH_3)_4Cl_2]^+$ is formed.

(i) What is the name of this complex ion? ..... (1 mark)

.....

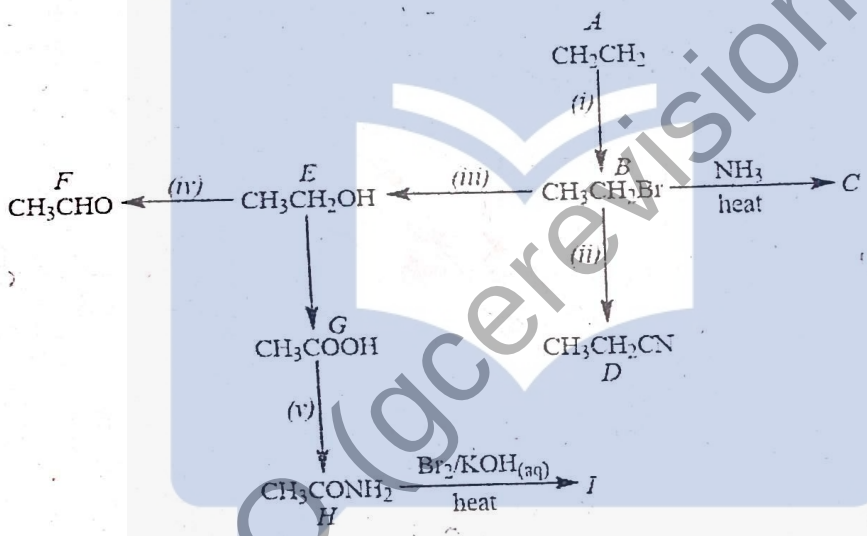
(ii) Draw the structures of the two geometric isomers of the complex ion. ..... (2 marks)

.....

(Total = 20 marks)

SECTION C: ORGANIC CHEMISTRY

5. (a) Study the following reaction scheme and answer the questions that follow



(i) Write the names and structural formulae of compounds *C* and *I*.

Compound *C*:

.....

Compound *I*:

..... (4 marks)

(ii) Give the reagents and reaction conditions necessary for the following conversions.

Steps	Reagents	Reaction Conditions
(i)		
(ii)		
(iii)		
(iv)		
(v)		

(5 marks)

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(iii) Compare the procedures for converting E to F and E to G

.....

.....(2 marks)

b) Suggest a chemical test to distinguish between:

(i) C_6H_5OH and $C_6H_5CH_2OH$

.....

.....

(ii) $CH_3CH_2NH_2$ and CH_3CONH_2

.....

.....

(iii) CH_3COCH_3 and $CH_3CO_2CH_3$

.....

.....(9 marks)

(Total = 20 marks)

6.(a) Give the formulas of two organic compounds that would react together to give each of the following products.

(i) CH_3COOCH_3

.....

.....

(ii) $CH_3CH_2NHCOCH_3$

.....

..... (4 marks)

(b) Provide the systematic names for the following amines.

(i) CH_3NHCH_3 .....

(ii) $C_6H_5CH_2NH_2$ .....

(iii) $C_6H_5NHCH_3$ .....

(iv). $C_6H_5NH_2$.....

(c) State and explain which of the two amines (phenylmethyl) amine and phenylamine is a weaker base.

State .....

Explanation .....

.....

.....(3 marks)

(d) Give the reagents and reaction conditions for the conversion of

(i) $CH_3CH(Br)CH_3$ to $CH_3CH(OH)CH_3$

Reagent: .....

Reaction condition .....

(ii) $CH_3CHBrCH_3$ to $CH_3CH\ CH_2$

Reagent: .....

Reaction condition: .....(2 marks)

(iii) Name the type of reaction mechanism taking place and state the role of the reagent in the formation of the product in d(ii) above.

Type of reaction mechanism: .....

Role of the reagent: .....(2 marks)

(iv) Outline the mechanism for the conversion of $CH_3CH(Br)CH_3$ to $CH_3CH(OH)CH_3$

.....

.....(2 marks)

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(e) Acylation of the benzene ring can occur with an acyl chloride by the reaction.

$$C_6H_6 + CH_3COCl \xrightarrow{AlCl_3} Y + HCl$$

(i) Draw the structure of the product Y.

.....

.....

(ii) State the type of organic reaction mechanism taking place in the above reaction and draw the structure of the electrophilic reagent in the acylation of benzene.

Name of organic reaction mechanism:

.....

Structure of electrophilic reagent:

.....

..... (3 marks)

(Total = 20 marks)

END

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